

# Rohit Gupta

1520 Eastlake Avenue E Apt 411, Seattle, WA 98102

☎ (+1) 4079905985

✉ rohitgupta.hp@gmail.com

🏠 rohit-gupta.github.io

📁 rohit-gupta

🔗 0WukQpMAAAAJ

📺 rohitguptahpf

## Education

- 2019 - 2025 **PhD. in Computer Science**, University of Central Florida, Orlando
- 2015 - 2017 **M.Tech. in Computer Science and Engineering**, IIT Kanpur, Kanpur, India
- 2010 - 2014 **B.Tech. in Electrical Engineering**, IIT Kanpur, Kanpur, India

## Selected Publications

|                                 |                                                                                                                                                                                                                                                                                                                                                                                                                 |
|---------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ACL 2026<br>Oral (Top 5%)       | <b>ViLLE: Video LLM Embeddings for Retrieval</b><br><b>R Gupta</b> , J Unnikrishnan, F Fei, S Liu, S Tran, M Shah<br>Pre-Print: <a href="https://arxiv.org/abs/2604.12148">https://arxiv.org/abs/2604.12148</a>                                                                                                                                                                                                 |
| CVPR 2026<br>Highlight          | <b>VRR-QA: Visual Relational Reasoning in Videos Beyond Explicit Cues</b><br>Sirnam S, <b>R Gupta</b> , P Kulkarni, D Shatwell, J Santiago, N Siddiqui, J Fiorese, M Shah<br>Link: <a href="https://openaccess.thecvf.com/content/CVPR2026/html/Swetha_VRR-QA">https://openaccess.thecvf.com/content/CVPR2026/html/Swetha_VRR-QA</a>                                                                            |
| CVPR 2026                       | <b>VidTAG: Temporally Aligned Video to GPS Geolocalization with Denoising Sequence Prediction at a Global Scale</b><br>PP Kulkarni, <b>R Gupta</b> , M Shah<br>Link: <a href="https://openaccess.thecvf.com/content/CVPR2026/html/Kulkarni_VidTAG">https://openaccess.thecvf.com/content/CVPR2026/html/Kulkarni_VidTAG</a>                                                                                      |
| ECCV 2026                       | <b>The Telephone Game: Evaluating Semantic Drift in Unified Models.</b><br>S Mollah, <b>R Gupta</b> , Sirnam S, Q Liu, A Munir, M Shah<br>arXiv: <a href="https://arxiv.org/abs/2509.04438">https://arxiv.org/abs/2509.04438</a>                                                                                                                                                                                |
| ECCV 2026                       | <b>SB-Bench: Stereotype Bias Benchmark for Large Multimodal Models.</b><br>V Narnaware, A Vayani, <b>R Gupta</b> , Sirnam S, M Shah<br>arXiv: <a href="https://arxiv.org/abs/2502.08779">https://arxiv.org/abs/2502.08779</a>                                                                                                                                                                                   |
| ECCV 2024<br>Citations: 11      | <b>Open Vocabulary Multi-Label Video Classification</b><br><b>R Gupta</b> , MN Rizve, A Tawari, J Unnikrishnan, S Tran, M Shah<br>(Link: <a href="https://www.ecva.net/papers/eccv_2024/papers_ECCV/papers/05599.pdf">https://www.ecva.net/papers/eccv_2024/papers_ECCV/papers/05599.pdf</a> )                                                                                                                  |
| CVPR 2023<br>Citations: 41      | <b>Class Prototypes based Contrastive Learning for Classifying Multi-Label and Fine-Grained Educational Videos</b><br><b>R Gupta</b> , A Roy, S Kim, C Christensen, T Grindal, S Gerard, M Cincebeaux, A Divakaran, M Shah<br>Link: <a href="https://openaccess.thecvf.com/content/CVPR2023/html/Gupta_Class_Prototypes_...">https://openaccess.thecvf.com/content/CVPR2023/html/Gupta_Class_Prototypes_...</a> |
| AAAI 2023<br>Citations: 15      | <b>Contrastive Self-Supervised Learning Leads to Higher Adversarial Susceptibility</b><br><b>R Gupta</b> , N Akhtar, A Mian, M Shah<br>Link: <a href="https://ojs.aaai.org/index.php/AAAI/article/view/26733">https://ojs.aaai.org/index.php/AAAI/article/view/26733</a>                                                                                                                                        |
| WACV 2026<br>Citations: 16      | <b>GAEA: A Geolocation Aware Conversational Assistant</b><br>R Campos, A Vayani, PP Kulkarni, <b>R Gupta</b> , A Dutta M Shah<br>Link: <a href="https://openaccess.thecvf.com/content/WACV2026/html/Campos_GAEA">https://openaccess.thecvf.com/content/WACV2026/html/Campos_GAEA</a>                                                                                                                            |
| ICPR 2020<br>Citations: 152     | <b>RescueNet: Joint building segmentation and damage assessment from satellite imagery</b><br><b>R Gupta</b> , M Shah<br>Link: <a href="https://ieeexplore.ieee.org/document/9412295">https://ieeexplore.ieee.org/document/9412295</a>                                                                                                                                                                          |
| CVIU, Jun '22<br>Citations: 276 | <b>TCLR: Temporal Contrastive Learning for Video Representation</b><br>I Dave, <b>R Gupta</b> , M N Rizve, M Shah<br>Link: <a href="https://www.sciencedirect.com/science/article/pii/S1077314222000376">https://www.sciencedirect.com/science/article/pii/S1077314222000376</a>                                                                                                                                |
| MediaEval 2018<br>Citations: 18 | <b>Linear Models for Video Memorability Prediction Using Visual and Semantic Features</b><br><b>R Gupta</b> , K Motwani<br>Link: <a href="http://ceur-ws.org/Vol-2283/MediaEval_18_paper_31.pdf">http://ceur-ws.org/Vol-2283/MediaEval_18_paper_31.pdf</a>                                                                                                                                                      |

## Pre-Prints

|              |                                                                                                                                                                                                                                                                                                                             |
|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Under Review | <b>BrailleVision: Text Instruction Tuning of LLMs to Improve Visual Skills</b><br><b>R Gupta</b> , MN Rizve, P Tirupattur, M Shah<br>(Private Pre-Print (Google Drive): <a href="https://drive.google.com/file/d/1XLrg3nzotRLJMq6IYpUpj8E3xsa1daxr">https://drive.google.com/file/d/1XLrg3nzotRLJMq6IYpUpj8E3xsa1daxr</a> ) |
| Under Review | <b>Cross-View Open-Vocabulary Object Detection in Aerial Imagery</b><br>J Kini, <b>R Gupta</b> , M Shah<br>Pre-Print: <a href="https://openreview.net/forum?id=QgTA3AvUC5">https://openreview.net/forum?id=QgTA3AvUC5</a>                                                                                                   |
| Under Review | <b>StretchySnake: Flexible VideoMamba for Short and Long-Form Action Recognition</b><br>N Siddiqui, <b>R Gupta</b> , Sirnam S, M Shah<br>OpenReview: <a href="https://openreview.net/forum?id=SMIVEoSyl">https://openreview.net/forum?id=SMIVEoSyl</a>                                                                      |
| Under Review | <b>Self-Supervision is Not All You Need: In Defense of Semi-Supervised Learning.</b><br><b>R Gupta</b> , MN Rizve, Sirnam S, N Kardan, M Shah<br>OpenReview: <a href="https://openreview.net/forum?id=nnYsWoe1ST">https://openreview.net/forum?id=nnYsWoe1ST</a>                                                            |

## Work Experience

### Applied Scientist II (L5), Amazon Prime Video Science

August 2025-Present,  
Seattle

- Developed a fast broad catalog video clips search engine and a highly adaptive state of the art within video clip search LLM agent.
- Search work is being leveraged by Amazon studios editors resulting in \$20 million worth of projected savings.
- Contributed to a hybrid audio-visual model for vending suitable cuepoints for Ads, increasing ad inventory by \$70 million annually.
- Built eval and training datasets for a text embedding model specialized for semantic matching of movie and TV show titles.
- Carried out embedding model training on 1 million+ titles, achieving upto 15% improvement in recommendation tasks.

Applied Scientist Intern, Amazon Nile, Rufus Multi-Modal Team

May-Nov 2024, Seattle

- Developed a single model capable of solving a wide spectrum of video understanding tasks: Question Answering, Video Retrieval, Localization etc
- Built an unified Video LLM with the capabilities of outputting text as well as embeddings in order to solve all the tasks.
- Carried out medium scale distributed post-training on 10 million+ videos.

Applied Scientist Intern, Amazon Search Science and AI, M5 Team

May-Nov 2023, Palo Alto

- Worked on multi-label open vocabulary video classification; recognizing objects and actions in videos not present in training data.
- Leveraged LLMs to assist CLIP Text Encoder by providing class attributes and designed Temporal Modeling adapters for the CLIP Vision Encoder
- Achieved competitive results across multiple video classification benchmarks in zero-shot open vocabulary setting.
- Work resulted in a successful publication at ECCV 2024 main conference.

Research Intern, SRI International

May- Aug 2022, Menlo Park (remote)

- Was tasked with solving the problem of classifying youtube education videos into fine-grained topic categories within subjects.
- Developed multi-label, multi-modal prototype contrastive learning to solve fine-grained video content understanding.
- Work resulted in a successful poster publication at CVPR 2023 main conference, and was invited as an Oral presentation at a workshop.

Research Engineer, Conduent Labs (erstwhile Xerox Research)

Sep. 2017- Jul. 2019, Bangalore, India

- Contributed to a variety of projects in Computer Vision: Video memorability prediction, Analyzing multi-modal data for smart-city applications, Instance recognition and image classification for augmented reality (AR) and appearance based re-identification of cars for traffic flow analysis.

Data Scientist, Fuzzy Logix

Jul. 2014 - Jun. 2015, Bangalore, India

- Developer on DB Lytix™ suite of machine learning, statistical and financial algorithms embedded into data warehouses like Teradata™& Netezza™

Achievements

|      |                                                                                            |
|------|--------------------------------------------------------------------------------------------|
| 2023 | <b>Oral Presentation</b> , Workshop on Large Scale Holistic Video Understanding, CVPR      |
| 2021 | <b>1st Place and Jury Prize</b> , VI-Priors Action Recognition Challenge, ICCV             |
| 2019 | <b>Fellowship</b> , ORCGS Doctoral Fellowship, UCF                                         |
| 2018 | <b>1st Place</b> , MediaEval 2018: Predicting Media Memorability Task                      |
| 2010 | <b>National Rank 433 (Top 0.1%)</b> , Joint Entrance Exam, Indian Institutes of Technology |

Service

|                  |                                                                                                |            |
|------------------|------------------------------------------------------------------------------------------------|------------|
| 2025,2026        | <b>Organizer</b> , Workshop on Video Large Language Models, CVPR                               |            |
| 2022-present     | <b>Reviewer</b> , CVPR, ECCV, ICCV, AAAI, ICLR, WACV, IEEE Journals (TPAMI, TIP, TNNLS, TCSVT) |            |
| 2020, 2022, 2025 | <b>Mentor</b> , NSF Research Experiences for Undergraduates, UCF-CRCV REU Site                 |            |
| 2015-16          | <b>Teaching Assistant</b> , Courses: Fundamentals of Computing, Machine Learning               | IIT Kanpur |

Research Overview

Agentic LLMs for Search

WORK PROJECT

Amazon ongoing

- Developing a variety of Agentic approaches for Prime Video use cases like title search, clip search, compelling clips selection etc.

Joint Generative and Embedding Video LLM

INTERNSHIP AND GRADUATE RESEARCH PROJECT

Amazon 2024

- Developed a multi-modal large language model (LLM) capable of generating both embeddings and text to solve video understanding tasks.

Text Instruction Tuning of LLMs for Better Visual Understanding

GRADUATE RESEARCH AS LEAD RESEARCHER

UCF 2024

- Designed a lightweight, specialized text instruction tuning stage for LLMs to improve their downstream multi-modal understanding performance.

Multi-Label Open Vocabulary Video Classification

INTERNSHIP AND GRADUATE RESEARCH PROJECT

Amazon 2023

- Developed a video classification model capable of recognizing object and action classes not seen during training.

Multi-Label Contrastive Learning for Fine-Grained Educational Video Classification

INTERNSHIP AND GRADUATE RESEARCH PROJECT

SRI & UCF 2022

- Achieved state of the art results on a novel dataset of education videos and two prior benchmark datasets (YouTube-8M and COIN)

Conversational Geolocalization Aware Assistant

GRADUATE RESEARCH AS COLLABORATOR

UCF 2025

- Helped design a dataset and finetune a multi-modal LLM to create a geolocalization expert conversational assistant for tourism.

Benchmarking Multi-Modal LLMs

GRADUATE RESEARCH AS COLLABORATOR

UCF 2025

- **SB-Bench**: Benchmark to identify if models end up relying on stereotypes in situations with limited information.
- **VRR-QA**: Benchmark to comprehensively test the video understanding capabilities of Video LLMs based on logical operators.
- **ImplicitQA**: Benchmark the video understanding capabilities of Video LLMs going beyond single frames and requiring complex reasoning.
- **ALM**: Benchmarking Multi-Modal LLMs to comprehensively test the multi-lingual and cross-cultural capabilities across 100 languages.

Temporal Contrastive Learning of Video Representations

GRADUATE RESEARCH AS COLLABORATOR

UCF 2021

- Devised a SotA self-supervised contrastive learning approach for video classification by enforcing temporal distinctiveness in feature space.